Sri Lanka Institute of Information Technology
BSc. (Hons) in Computer Systems Engineering

Control Theory
IE2074
Final **individual** Assignment

The Project Brief

Design and Implementation of PID-Based DC Motor Speed Control using ESP32 and L298N Motor Driver

Objective

- Understand the concept of PID control
- Learn PWM motor control using ESP32
- Interface L298N motor driver
- Implement closed-loop control system
- Analyze system response and tune PID parameters

Required Components

- ESP32 Development Board
- L298N Motor Driver Module
- DC Motor (with encoder preferred)
- Connecting wires
- Breadboard

Assignment Guidelines

1. Deadline for the report: 27th April 2026
2. File Naming Convention: Final report must follow: StudentNum_IE2074_Assignment.pdf

Report Requirements - Must Include Sections

- Working principle (with Mathematical representation)
- Circuit diagram
- Source code (Arduino ide with C/C++)
- PID tuning values
- Graphs (Speed vs Time)
- Challenges faced
- Improvements

Report Format

1. Cover Page
    - Course / Module name
    - Lab Assignment Title:
    - Group Number:
    - Student IDs and Names:
2. Introduction
3. Design Methodology
4. Component Selection
5. Results & Testing
6. Discussion
7. Conclusion
8. References